Causal Identity in Oriented Time

How Descriptive Frames Partition Processes into Entities, Events, Causes, and Effects

Enumeration, Closure, Factorization, Invariance, Irreducibility, and Residue Across Varied Axioms

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Documentation and Scope Disclosure

This laboratory note records an independently developed line of inquiry within the Lambda / NSAF / TUFT and Registry Studies research programs. It does not claim ownership, invention, discovery, or priority over Euler's formula or identity, Sophie Germain's factorization, Noether's theorems, prime factorization, p-adic analysis, analytic number theory, relativity, evolutionary theory, causal modeling, proof enumeration, category theory, obstruction theory, or any other established result, field, or terminology discussed here.

The proposed contribution is comparative, architectural, and interpretive. It asks how a descriptive frame partitions an ongoing process into entities, events, temporal orders, causes, and effects; how exact mathematical operations can serve as controlled exemplars of different partitioning disciplines; and how residue may be recorded when one causal description does not preserve all distinctions carried by another.

Throughout, established mathematics, framework definitions, derived observations, modeling interpretations, and conjectural extensions are marked separately. This document is an independent progress record rather than a claim of completed universal theoremhood.

Abstract

Causal explanations often appear to report relations among already identifiable entities arranged in an independently given temporal order. This paper investigates a prior layer of construction: the frame-relative operations through which an ongoing process becomes divided into entities, events, temporal intervals, causes, and effects. A descriptive-causal frame selects a domain, grain, admissible distinctions, continuity criteria, event boundaries, temporal orientation, causal rules, and observation maps. Different frames may therefore generate non-isomorphic causal graphs over a shared or overlapping process record without immediately producing contradiction.

Several established mathematical structures serve as controlled exemplars rather than as competing universal definitions of identity. Polygonal convergence toward circular circumference illustrates finite linear refinement approaching curved closure. Exponential limits illustrate discrete iteration approaching continuous proportional flow. Euler's identity exemplifies completed closure within the complex exponential frame. Sophie Germain's identity exemplifies exact decomposition and recomposition through oppositely signed mixed terms. Noether's theorem exemplifies persistence through invariance under declared transformations. Prime numbers exemplify irreducibility under integer factor reproduction, while the corrected registry account distinguishes abstract primality from base- or modulus-dependent artifacts of visibility. Infinite-registry constructions add stage, provenance, search history, and positional recoverability to the causal account.

The paper then treats time as oriented traversal rather than mere enumeration. A sequence of samples does not alone determine an entity, a causal arrow, or an explanation. Identity bindings determine which manifestations count as one continuing subject; event partitions determine which changes become causal nodes; and temporal and propagation rules determine which directed relations remain admissible. At coarser narrative grains, distributed constraints and attractor structures may become compressed into singular agents, purposes, or acts of will.

The Lambda Principle is used as a residue-accounting method for translations among causal framings. A translation may preserve observations or invariants while failing to preserve entity boundaries, event structure, causal depth, temporal grain, or interpretation. Such residue does not automatically establish the superiority of one account or the falsity of another. It identifies the boundary at which one causal description ceases to reproduce distinctions carried by another.

Claim-Status Legend

Established mathematics	Standard results stated under their ordinary hypotheses. These statements do not depend upon the proposed framework.
Framework definition	Vocabulary and formal objects introduced here to make the comparison precise.
Derived observation	A consequence of established mathematics after adopting the framework definitions.
Modeling interpretation	A proposed structural reading that requires an explicit model before it can become theorem-level mathematics or physics.
Open program	A conjecture, engineering proposal, or research direction not asserted as established.

Contents

1 Orientation: descriptions encode axioms

Governing Thesis

A description does not enter a world empty-handed. It brings distinctions, permissible operations, continuity criteria, scales, temporal orientations, and standards of evidence. These commitments function as axioms of description. They determine which manifestations receive binding as one entity, which differences count as events, which sequences count as histories, and which relations qualify as causes.

A causal account therefore contains more than an observed process. It contains a framing discipline through which that process becomes divided, ordered, measured, and narrated. The framework proposed here does not deny physical causality or mathematical objectivity. It asks a prior question: what descriptive commitments make a particular causal assertion well formed?

The surface grammar

$$A \text{ caused } B$$

usually presupposes at least six prior commitments:

1. a rule that identifies A;
2. a rule that identifies B;
3. a criterion by which A and B persist through change;
4. an event partition separating "before" from "after";
5. an ordering relation placing the relevant events in sequence;
6. a causal rule determining which ordered relations count as influence, production, constraint, or evidence.

These commitments may be empirically disciplined, mathematically exact, narratively authored, or some combination of the three. Their existence does not make causality arbitrary. It makes causal description auditable.

1.1 Map, marker, and transfer

The Primitive Semantic Registry Protocol frames a descriptive system as a map-making structure: a marker, frame, operation, measurement whole, admissible transformations, invariants, and residues (Semita 2026c). This paper adopts that discipline and applies it specifically to causation.

A map can be exact within its domain without becoming identical to the domain. A clock can divide duration without exhausting time. A coordinate chart can support exact calculation without becoming the manifold. A numeral base can display divisibility without creating primality. A causal graph can encode dependencies without becoming the process in every respect.

The governing operational rule is:

> Keep the marker. Trace the operation. Watch the invariant. Record the residue. Transfer only what survives.

1.2 A generative order of causal description

The paper uses the following sequence as its main architectural spine:

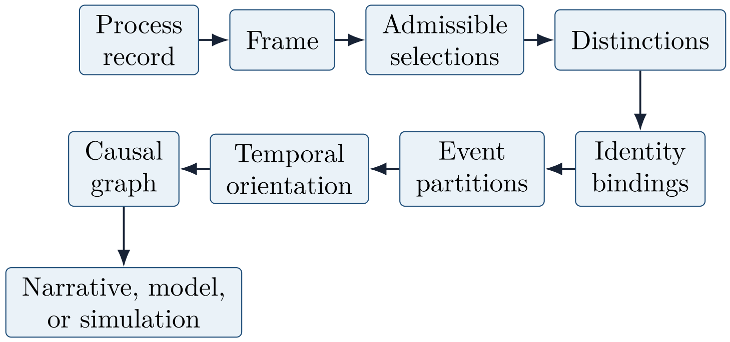

Figure 1: The proposed generative order. Cause does not enter first; it appears after selection, identity binding, event partition, and temporal orientation.

The arrows in figure 1 should not be read as a metaphysical claim that reality is produced by language. They describe the dependency structure of an articulated causal account.

2 The descriptive-causal frame

Definition 2.1 (Process record)**.** *A* process record $\mathcal{P}$ *is any declared collection of states, observations, events, relations, samples, measurements, or histories used as the carrier for a causal description. It may be continuous, discrete, sampled, partially observed, or multiply represented.*

Definition 2.2 (Descriptive-causal frame)**.** *A descriptive-causal frame is a tuple*

$$\mathfrak{F} = (\Omega, G, \Sigma, B, E, \prec, C, O),$$

where:

- Ω *is the admitted domain or carrier;*
- G *is the descriptive grain;*
- Σ *is the family of admissible selections or distinctions;*
- B *is an identity-binding rule;*
- E *partitions selected change into events;*
- $\prec$ *is a temporal, logical, or procedural ordering relation;*
- C *is the admissible causal relation;*
- O *is the observation or registration map.*

This tuple does not assert that every causal theory uses the same ontology. It supplies a comparison scaffold. A relativistic field theory, a discrete-event simulation, a biological population model, a fictional narrative, and a proof search can instantiate its components differently.

Definition 2.3 (Frame-generated causal graph). *Given a process record $\mathcal{P}$ and frame $\mathfrak{F}$, let*

$$\mathcal{G}_{\mathfrak{F}}(\mathcal{P})$$

denote the directed graph whose nodes are events admitted by E and whose directed edges are relations admitted by C under the ordering $\prec$.

The notation resembles the graphical language of causal modeling, but the present use remains broader than a structural causal model in the Pearlian sense (Pearl 2009). No conditional-independence semantics, intervention calculus, or probability factorization should be inferred unless a branch explicitly supplies them.

Proposition 2.4 (Frame-relative causal partition). *Two descriptive frames may generate non-isomorphic causal graphs over a shared or overlapping process record because they differ in grain, selection, identity binding, event partition, temporal order, observation, or admissible causal relation. Non-isomorphism alone does not establish contradiction.*

Framework justification. Let $\mathfrak{F}$ and $\mathfrak{F}'$ differ in at least one of $G, \Sigma, B, E, \prec, C, O$. If B binds manifestations differently, the node-bearing entities differ. If E partitions the record differently, the event-node set differs. If $\prec$ or C differs, the edge relation differs. Therefore the resulting graphs need not be isomorphic. Contradiction requires more than non-isomorphism: the graphs must make incompatible assertions under sufficiently shared meanings, scope, identity criteria, temporal rules, and comparison conditions. □

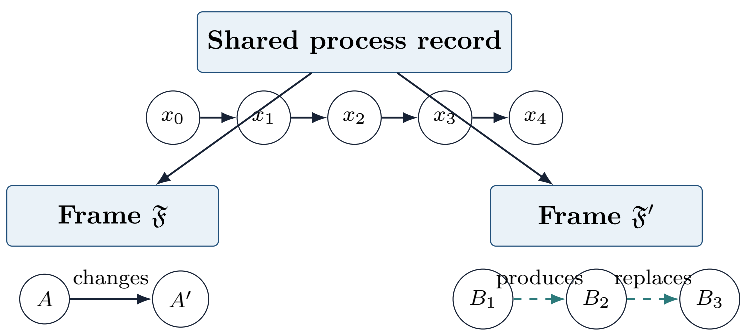

Figure 2: One frame binds the record as the transformation of a continuing entity; another partitions it into production or replacement events among distinct entities.

2.1 Contradiction and frame collision

A *frame collision* occurs when a description combines commitments drawn from different frames without declaring a bridge. A contradiction requires stronger conditions:

1. the same terms retain the same meanings;
2. the same frame and scope apply;
3. the same temporal and identity rules apply;
4. the claims assert mutually incompatible outcomes.

This distinction prevents two opposite errors. It prevents premature dismissal of a description whose difference arises from grain or scope, and it prevents rhetoric about framing from dissolving genuine logical or empirical contradiction.

3 Enumeration, continuity, and oriented time

3.1 Polygonal refinement and curved closure

For a circle of radius R, the perimeter of a regular inscribed n-gon is

$$P_n = 2nR\sin\left(\frac{\pi}{n}\right).$$

As $n \to \infty$,

$$\lim_{n\to\infty} P_n = 2\pi R.$$

Established Mathematics

At every finite n, the inscribed polygon remains a polygon, not a circle. The sequence converges to the circumference in the ordinary analytic sense. Calling the limit an "infinitely sided polygon" may offer intuition, but it should not be treated as an ordinary polygon with an actually completed infinite list of sides.

The causal relevance lies in the relation between isolated samples and ordered refinement. No single finite polygon contains the full curved circumference. The limiting structure becomes available through the sequence and its convergence law. Thus the identity of a continuous target may be disclosed by an ordered family of finite distinctions without being identical to any member of that family.

3.2 Discrete iteration and continuous proportional change

The classical limit

$$e^{rt} = \lim_{n\to\infty}\left(1 + \frac{rt}{n}\right)^n$$

relates repeated finite compounding to continuous proportional growth. The differential equation

$$\frac{dx}{dt} = rx$$

has the solution

$$x(t) = x_0 e^{rt}.$$

At finite grain, a registry may store values $x_0, x_1, x_2, \ldots$. The curve appears through the relation among values, the update rule, and the parameter ordering. An isolated sample contains a state; it does not contain the entire process.

3.3 Euler's formula and orientation

Euler's formula states

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

At $\theta = \pi$,

$$e^{i\pi} + 1 = 0.$$

These identities are exact within complex analysis (National Institute of Standards and Technology 2026). Yet both $e^{i\theta}$ and $e^{-i\theta}$ remain available. They traverse the same unit circle with opposite orientation.

Framework Definition

Neither enumeration, exponential structure, nor circular closure alone chooses a physical future. A causal arrow requires additional structure: initial and boundary conditions, permitted successors, finite propagation, record formation, dissipation, coarse-graining, inaccessible inverses, or an observer-relative registry orientation.

3.4 Seven distinguishable temporal roles

Temporal role	Function in a description
Enumerative time	Orders records: $0, 1, 2, \ldots$. It supplies position in a sequence without necessarily supplying duration or causation.
Metric time	Measures intervals: $\Delta t = t_{n+1} - t_n$.
Dynamical time	Orients an update law: $x_{n+1} = T(x_n)$ or $\dot{x} = F(x, t)$.
Causal time	Orders influence or dependency: $e_i \prec_C e_j$.
Propagation time	Measures travel through a channel: $T_{\text{prop}} = d/v$ in the constant-speed case.
Registry time	Records when evidence, descriptions, witnesses, or explanations become available to a system.
Narrative time	Records the order in which events receive presentation, recollection, or interpretation.

These roles can align without becoming identical. A flashback reverses narrative order but not physical propagation. A time-reversal-invariant equation may coexist with asymmetric boundary conditions. A cause may occur before a registry can detect or certify it. A sampled worldline may be rendered continuously even though only discrete samples were stored.

4 Identity binding and the causal subject

Definition 4.1 (Identity binder)**.** *Given selected manifestations $x_0, \ldots, x_n$ from a process record, an identity binder $B_{\mathfrak{F}}$ assigns them to one or more entity histories according to the continuity criteria of frame $\mathfrak{F}$.*

A frame may bind

$$B_{\mathfrak{F}}(x_0, x_1, \ldots, x_n) = A,$$

while another frame may bind

$$B_{\mathfrak{F}'}(x_0, x_1) = A_1, \quad B_{\mathfrak{F}'}(x_2, x_3) = A_2, \quad B_{\mathfrak{F}'}(x_4, \ldots, x_n) = A_3.$$

Under the first frame, the causal sentence may read: "A changed." Under the second: "A_1 produced, became, or was replaced by A_2." The process record need not change for the causal grammar to change.

4.1 Identity as invariant signature

A useful framework-level representation is an identity discipline

$$\mathfrak{D} = (\mathcal{R}, \mathcal{A}, \mathcal{I}, \mathcal{E}),$$

where $\mathcal{R}$ is a registry, $\mathcal{A}$ a family of admissible transformations, $\mathcal{I}$ a family of invariants, and $\mathcal{E}$ an equivalence rule.

For $x \in \mathcal{R}$, define the invariant signature

$$\mathrm{Id}_{\mathfrak{D}}(x) = \{I \in \mathcal{I} : I(a(x)) = I(x) \text{ for every applicable } a \in \mathcal{A}\}.$$

Modeling Interpretation

This definition does not claim that identity universally reduces to a set of invariants. It supplies a disciplined way to state what a selected frame requires to persist. Material continuity, structural continuity, legal continuity, functional continuity, memory continuity, and lineage continuity may yield different equivalence relations.

4.2 Nouns as compressed histories

A noun-like description selects a stabilized manifestation: "the particle," "the organism," "the hammer," "the species." A process description foregrounds transformations and relations. Neither grammatical form is inherently false, but each exposes and suppresses different causal structure.

An entity history can be represented as

$$\gamma_E = (x_0, T_0, x_1, T_1, x_2, \ldots),$$

with the frame specifying which transformations preserve identity. The entity becomes a bound trajectory through state space rather than a timeless point.

5 Mathematical exemplars of causal organization

The examples in this section do not form an exhaustive taxonomy of identity. They demonstrate how exact mathematical operations can alter what appears as a causal unit.

5.1 Eulerian closure: unity and causal compression

Euler's identity

$$e^{i\pi} + 1 = 0$$

exemplifies completed closure. It foregrounds exact relation, unity, and internal consistency. It does not, by itself, specify a generative history, causal direction, decomposition, or transition map. The Registry Factorization Principle therefore treats Euler as a closure exemplar rather than as universal transition machinery (Semita 2026d).

a completed identity can compress the path by which its terms became related. When imported into causal narrative, closure may be mistaken for explanation. "These terms close" does not automatically mean "this closure generated the process."

5.2 Germain factorization: relational decomposition and exact reattachment

Sophie Germain's identity is

$$x^4 + 4y^4 = (x^2 + 2xy + 2y^2)(x^2 - 2xy + 2y^2).$$

Let

$$A = x^2 + 2y^2, \qquad B = 2xy.$$

Then

$$A^2 - B^2 = (A + B)(A - B),$$

and therefore the factorization follows exactly.

Define

$$Q = x^4 + 4y^4, \qquad F_\pm = x^2 \pm 2xy + 2y^2.$$

Then

$$Q = F_+ F_-.$$

Established Mathematics

The algebraic factorization and exact recomposition require no approximation, limiting process, or physical assumption.

Modeling Interpretation

Under a registry reading, the common quartic acts as a lifted container; the opposite mixed terms expose paired coupling; multiplication reattaches the factors and reconstructs the invariant. This is a disciplined algebraic model of "lift - twist - reattach." It is not, without additional structure, a physical equation of motion, relativistic coordinate transformation, manifold atlas, or universal observer theory.

The causal lesson is that what appears as one completed entity in one frame may appear as a relation among coupled components in another.

5.3 Noether invariance: persistence through transformation

Consider a smooth Lagrangian system with action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\, dt.$$

Suppose a one-parameter transformation

$$\delta q^i = \varepsilon \xi^i(q, t), \qquad \delta t = \varepsilon \tau(q, t)$$

changes the Lagrangian by a total derivative,

$$\delta L = \varepsilon \frac{dF}{dt}.$$

Then, along Euler-Lagrange solutions, the Noether quantity

$$Q_N = \sum_i p_i \xi^i - H\tau - F$$

is conserved under the ordinary regularity assumptions, where $p_i = \partial L/\partial \dot{q}^i$ and $H = \sum_i p_i \dot{q}^i - L$ (Noether 1971; Kosmann-Schwarzbach 2020).

Established Mathematics

Noether's theorem establishes a precise relationship between continuous variational symmetries and conservation laws under stated hypotheses. It does not say that every persistence relation in every domain is Noetherian.

The causal lesson is that the continuing subject of a history need not remain pointwise unchanged. Identity may be assigned through an invariant preserved across transformation.

5.4 Prime irreducibility: explanatory stopping under a declared operation

For $p \in \mathbb{N}$ with $p > 1$,

$$p \text{ is prime} \iff p = ab \Longrightarrow a = 1 \text{ or } b = 1$$

for $a, b \in \mathbb{N}$.

Primality is therefore irreducibility relative to integer multiplication. It does not imply irreducibility under every operation or in every enlarged algebraic domain.

The early Rope-and-Sand language described primes as grains that resist recursive subdivision. The corrected account is narrower and stronger:

> Primality is not built from the register. A register measures, displays, or foregrounds primality's relation to divisibility.

A base, modulus, grid, clock, or harmonic partition changes the visible artifact of divisibility without changing abstract primality (Semita 2026c).

For a nonzero rational number x, write

$$x = p^{v_p(x)} \frac{a}{b},$$

where $p \nmid a$ and $p \nmid b$. The p-adic norm is

$$|x|_p = p^{-v_p(x)}, \qquad |0|_p = 0.$$

Numbers become p-adically close when their difference is divisible by a high power of p (Gouvêa 1997).

The causal lesson is that a frame's primitive explanatory units depend upon the operations under which decomposition is admitted.

5.5 Infinite registry identity: stage, provenance, and recoverability

The Infinite Registry and Indexed-Residue Proof Engine distinguishes theoremhood, representation cost, and finite search state (Semita 2026e). Let $R(\varphi)$ be a declared proof representation class, K a trusted checker, and O a cost functional. Define

$$I_{R,c}(\varphi) = \inf \{O(p, \varphi; R, c) : p \in R(\varphi),\ K(\text{expand}(p)) = \text{accept}\}.$$

At finite enumeration stage N, define

$$b_N(\varphi; R, c) = \min \{O(p) : p \text{ appears by stage } N \text{ and verifies } \varphi\}.$$

The sequence b_N is nonincreasing, but $b_N = \infty$ at a finite stage does not imply $I_{R,c}(\varphi) = \infty$.

The registry develops through a directed family

$$P_0 \longrightarrow P_1 \longrightarrow P_2 \longrightarrow \cdots ,$$

where each stage preserves previous certified objects while enlarging the searchable registry.

The causal lesson is that occurrence, observability, discoverability, reconstruction, and certification may occupy different temporal positions. A causal route can exist in the process before a registry has found or represented it.

6 Causation must be typed

The verb "cause" often compresses several distinct relations. The framework therefore treats causal edges as typed.

Relation type	Operational meaning
Generative	Produces a successor state under a declared transition rule.
Propagative	Carries an influence, signal, disturbance, or constraint through a channel.
Enabling	Makes an event possible without uniquely producing it.
Constraining	Removes or suppresses possible continuations.
Conserving	Preserves an invariant through transformation.
Selective	Changes the relative continuation, survival, or reproduction of alternatives.
Attractive	Directs trajectories toward a region under explicitly defined dynamics.
Constitutive	Participates in what the entity or process is, rather than acting as an external trigger.
Intentional	Represents a goal held and acted upon by an agent under an agent model.
Retrospective	Summarizes a known outcome from a later vantage point.
Interpretive	Assigns meaning without necessarily acting as a physical force.
Certificational	Establishes a relation inside a proof, audit, or registry.

Typed edges allow a description to say that a boundary constrained a wave, a signal propagated to a detector, a conserved quantity persisted, an ecological relation selected among variants, and a later historian interpreted the outcome - without flattening all of these into one causal mechanism.

7 Physical causality as a rigorous exemplar

The physical causal-resolution layer supplies one strict rule pack within the broader descriptive environment. It does not define the whole environment.

7.1 A turn as a causal interval

For a propagation channel with speed v_{ch} and turn duration Δt,

$$d_{\max} = v_{\mathrm{ch}} \Delta t.$$

For electromagnetic propagation in vacuum, $v_{\mathrm{ch}} = c$. For sound, stress, fluid, or material waves, the applicable speed is generally lower and may depend on state or medium.

A turn contains:

- start time t_n and duration Δt_n;
- initial branch state;
- signals arriving during the interval;
- permitted local interactions;
- events generated during the interval;
- signals launched but not yet delivered;
- resulting state at t_{n+1}.

This structure, developed in the Previs Studio causal-resolution notes, treats enumeration as both interval index and physical duration (Semita 2026b).

7.2 Propagation edges and response latency

For a homogeneous channel across distance d,

$$T_{\text{prop}} = \frac{d}{v_{\text{ch}}}.$$

An effect becomes eligible only when

$$t_{\text{effect}} \geq t_{\text{cause}} + T_{\text{prop}} + T_{\text{response}}.$$

For variable speed along path γ,

$$T_{\text{prop}} = \int_{\gamma} \frac{ds}{v_{\text{ch}}(x, t, \sigma)}.$$

This rule requires signals in transit, delayed bindings, and explicit arrival events. It prevents a software update loop from silently turning every dependency into an instantaneous influence.

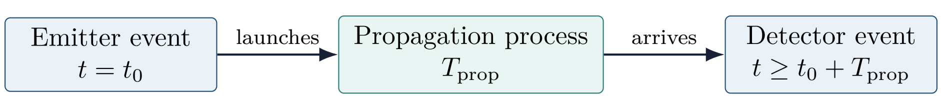

Figure 3: A causal edge contains duration and channel rules rather than an instantaneous link.

7.3 Causal depth and microsteps

A turn need not limit event count. It limits the causal depth that can fit within the interval. If A affects B, B responds, and then B emits toward C, the combined travel and response time must fit before C can respond. Otherwise the final influence remains in transit.

This prevents superluminal or infinite cascades inside one unsplit update. It also separates numerical substeps from visible narrative turns.

7.4 Particles, waves, fields, and measurements

A particle frame may store a sampled worldline with position, momentum, energy, internal state, received interactions, and emitted interactions. A massive particle satisfies

$$\|\Delta x\| < c\Delta t$$

in a relativistically constrained branch.

A wave or field frame stores distributed amplitude, phase, frequency, polarization, medium, source history, and boundary interactions. It advances the field according to its governing equation rather than translating one rigid mesh.

A measurement frame records discrete registrations. These frames may refer to one experimental preparation without being forced into one ontology. The environment should mark what was authored, calculated, sampled, interpolated, observed, inferred, predicted, or interpreted.

7.5 Relativistic scope

The baseline causal-turn model may use a preferred laboratory coordinate frame for computation. It should state that assumption openly. A full relativistic branch would add proper time, Lorentz transformations, frame-dependent simultaneity, invariant intervals, and model-specific causal cones. Einstein's 1905 treatment supplies the historical foundation for the relativity of simultaneity and invariant light-speed postulate (Einstein 1905).

8 Biological and narrative causality

8.1 The giraffe sentence as compressed causal architecture

The sentence

> The giraffe lengthened its neck because it needed to reach high leaves.

compresses many organisms, variation, inheritance, development, environmental distribution, feeding behavior, competition, survival, reproduction, and many generations into one grammatical subject: "the giraffe." It then converts a distributed environmental relation into an internal motive: "need." Finally, it lets a later function appear as the singular earlier cause of its own production.

The sentence may remain memorable and useful. Its causal status must nevertheless be distinguished from a forward-generative model.

8.2 A population-level model

Let $p_n(\ell)$ denote the distribution of a heritable trait ℓ in generation n. A schematic selection-and-transmission model is

$$p_{n+1}(\ell) = \frac{w(\ell, E_n) \int K(\ell \mid \ell') p_n(\ell')\, d\ell'}{Z_n},$$

where:

- $K(\ell \mid \ell')$ represents transmission, recombination, development, and variation;

- E_n represents relevant environmental relations;
- $w(\ell, E_n)$ represents differential contribution to later generations;
- Z_n normalizes the distribution.

High foliage can enter E_n and alter w without acting as a future intention that reaches backward to modify ancestors. Darwinian selection remains historical and population-level, while present function and past origin require separation (Darwin 1859; Mayr 1961; Gould and Lewontin 1979).

8.3 Six causal readings

1. **Individual action:** an organism reaches toward food.
2. **Developmental process:** growth and use occur under biological constraints.
3. **Population process:** trait distributions change across generations.
4. **Ecological constraint:** environmental relations alter available continuations.
5. **Attractor description:** trajectories tend toward a region under an explicit state space and update law.
6. **Teleological narrative:** the species developed a trait because it needed the eventual result.

The sixth may summarize function or significance, but it should not automatically inherit the causal status of the first five.

8.4 Attractor is not intention

An attractor is defined relative to a state space and dynamics. If T is a map on state space X, a set $A \subset X$ may act as an attractor when nearby trajectories approach A under iteration, subject to the relevant definition. The attractor does not thereby become an agent that anticipates or desires the limiting state.

Audit Condition

Whenever attractor language enters a biological or narrative explanation, the paper or software should declare the state variables, update rule, neighborhood, and convergence criterion. Otherwise "attractor" risks becoming a technical-sounding synonym for purpose.

8.5 Grain changes the apparent actor

At fine grain, mutations, developmental events, individual feedings, births, and deaths appear. At intermediate grain, organisms, populations, and niches appear. At coarse narrative grain, "the species," "nature," or "evolution" may appear as singular actors.

The central result is:

> Changing the entity grain changes who or what appears to act.

This does not mean that all grains are equally useful for every question. It means that a causal claim should disclose its grain.

9 Registry causality and epistemic time

A causal process can occur before it becomes observed, represented, reconstructed, or certified. The paper therefore distinguishes:

$$t_{\text{occurrence}}, \quad t_{\text{observation}}, \quad t_{\text{discovery}}, \quad t_{\text{reconstruction}}, \quad t_{\text{certification}}.$$

These times may coincide, but they need not.

The Infinite Registry sharpens this distinction. A finite stage may contain no proof or causal route even though a later stage discovers one. Search failure is therefore stage-relative evidence, not an oracle of global absence. The full registry preserves rejected and redundant routes as information about process, not as accepted mathematics (Semita 2026e).

This yields an epistemic arrow distinct from physical propagation. A later registry state can contain more causal structure than an earlier one without retroactively changing the event that occurred.

10 Cross-frame translation and Lambda causal residue

10.1 Bridges between causal graphs

Let

$$\Phi : \mathcal{G}_{\mathfrak{F}}(\mathcal{P}) \longrightarrow \mathcal{G}_{\mathfrak{F}'}(\mathcal{P})$$

be a proposed translation. It may preserve some nodes, edges, observations, invariants, or order relations while losing others.

Definition 10.1 (Causal residue register)**.** *A provisional causal residue register for translation Φ is*

$$\Lambda_\Phi = (R_B, R_E, R_\prec, R_C, R_O, R_I),$$

where the components record unmatched identity bindings, event partitions, temporal orders, causal relations, observations, and interpretations.

This tuple is an accounting schema, not a claim that residue is one universal substance.

10.2 Comparison statuses

A bridge may be:

- exact and invertible;
- many-to-one;
- approximate within tolerance;
- scale-limited;
- observation-equivalent;
- structurally analogous;
- interpretive only;

- unresolved.

Observation-equivalence does not establish ontological identity. Two models may predict the same detector event while disagreeing about intermediate localization, entity count, or hidden state.

10.3 Lambda as disciplined residue accounting

The corrected Lambda work does not present Lambda as one already-proved universal theorem. It presents a domain-sensitive schema: declare a frame of admissible operations, identify its reachable portion of a target, and classify the structured remainder with the established mathematics of that domain (Semita 2026a).

In the present paper, Lambda records what a causal translation fails to preserve. It does not automatically identify the source frame as true, the target as false, or the residue as a physical object.

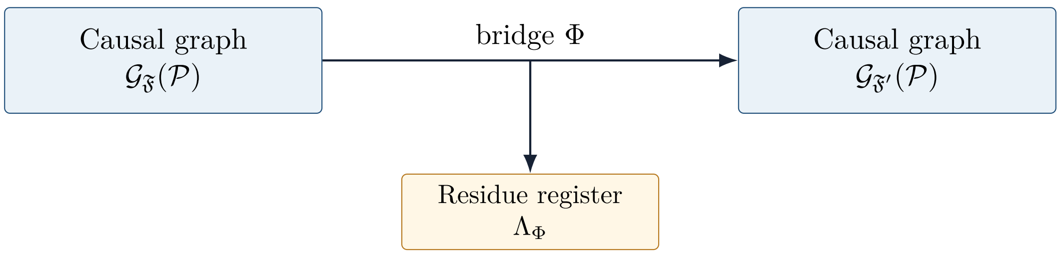

Figure 4: A bridge carries declared structure. The residue register records what fails to transfer without forcing either graph to become the universal measure of the other.

11 Scientific accountability and fictional freedom

Recognizing frame construction does not imply that all causal descriptions possess equal predictive, mathematical, or empirical standing. It distinguishes kinds of value rather than collapsing them.

Value domain	Typical accountability conditions
Scientific	Calibration, empirical support, prediction, reproducibility, uncertainty, and declared domain.
Mathematical	Definitions, axioms, proof, counterexample, scope, and dependency control.
Engineering	Operational performance, safety margin, failure modes, tolerances, and testability.
Narrative	Coherence, significance, character continuity, affect, symbolism, and authored intention.
Interpretive	Explanatory usefulness, perspective, historical fit, and acknowledged alternatives.

A fictional account may have greater value to a writer than a physically realizable account. A scientific user may instead want the exact boundary at which a fictional continuation departs from a selected physical rule set. The environment should support both without disguising one as the other.

The governing studio principle is:

> A description may be freely imagined. A relation may be rigorously governed. A frame declares where the rule applies. A branch preserves alternatives. A bridge permits interaction. A comparison reveals agreement and residue without forcing one account to become the measure of every other.

12 Design implications for a descriptive environment

12.1 Core primitives

The environment should support at least:

- identity;
- state;
- distinction;
- transformation;
- event;
- binding;
- boundary;
- propagation;
- turn;
- sample;
- observation;
- frame;
- branch;
- bridge;
- invariant;
- residue.

These should remain typed rather than collapse into a generic object class.

12.2 Epistemic status

Every rendered or reported item should carry one status:

- authored;
- calculated;
- sampled;
- interpolated;
- observed;

- inferred;
- predicted;
- interpreted;
- fictionally asserted;
- conflicted.

A smooth curve drawn between samples should not silently become an observed continuous trajectory. A detector event labeled "particle arrival" should retain the distinction between registration and interpretation.

12.3 Branching across varied axioms

A branch should fork whenever a consequential assumption changes:

- entity continuity;
- temporal grain;
- finite or instantaneous propagation;
- local or nonlocal interaction;
- particle or field state;
- deterministic or stochastic update;
- physical or narrative causation;
- one boundary condition or another.

The parent branch remains preserved. Integration should create a child branch with explicit translation contracts rather than overwrite the parents.

12.4 Causal audit

A branch audit should ask:

1. What identities exist, and by what continuity rule?
2. What grain defines an event?
3. Which temporal role applies?
4. What relation type supports each causal edge?
5. What propagation and response latency applies?
6. What was authored, calculated, observed, or inferred?
7. Which invariants must remain preserved?
8. What changes when the branch forks?
9. What residue remains under translation?

13 Falsifiers, limitations, and non-claims

The framework would be weakened if it could not distinguish its own definitions from empirical claims. Its central proposition is partly structural: changing node or edge construction can change a graph. Its substantive value therefore depends on whether the framework improves explanation, simulation, comparison, or error diagnosis.

13.1 Operational tests

The method should be judged by whether it can:

- identify hidden shifts of grain or entity binding in real causal disputes;
- prevent impossible propagation in simulation;
- distinguish narrative teleology from forward-generative models;
- preserve observation agreement while exposing ontological or interpretive differences;
- record translation loss more precisely than a binary compatible/incompatible label;
- generate reproducible branch comparisons from declared rule sets.

13.2 Explicit non-claims

This paper does not claim:

- that physical causality consists only of language;
- that every identity relation is arbitrary;
- that Euler, Germain, Noether, primes, and registries are mathematically identical;
- that Germain factorization supplies a physical transition law;
- that a base creates primality;
- that attractors possess intention;
- that every paradox dissolves through reframing;
- that a finite search failure proves global nonexistence;
- that Lambda replaces established obstruction theorems;
- that all descriptions have equal evidentiary status.

14 Research program

Open Program / Conjectural Frontier

The next formal steps are:

1. define a category of descriptive-causal frames with morphisms preserving declared subsets of structure;
2. distinguish exact, approximate, observational, and interpretive bridge classes;
3. define obstruction conditions for global reattachment of local causal descriptions;
4. implement a demonstrator that generates and compares causal graphs from alternative identity binders and temporal grains;
5. test the framework on bounded physical, biological, narrative, and proof-registry cases;
6. develop quantitative residue measures only where a domain supplies an explicit metric or obstruction functional.

A category-theoretic formulation would require objects, morphisms, preserved invariants, composition, identities, and possibly pullbacks or fibered constructions. No such universal category is claimed here. It remains an open program.

15 Conclusion

This work does not claim that causal structure consists only of framing or narration. It claims that every articulated causal account requires a discipline of distinction through which processes receive partition into entities, events, temporal orders, and relations.

Physical, mathematical, biological, computational, and fictional descriptions may impose different disciplines. Their differences should be inspected through declared axioms, preserved invariants, observations, and residues rather than concealed beneath apparently self-evident nouns and causal verbs.

The mathematical exemplars clarify distinct operations. Polygonal sequences show finite linear refinement approaching curved closure. Exponential limits show discrete iteration approaching continuous flow. Euler's identity shows exact closure. Germain's identity shows exact decomposition and recomposition. Noether's theorem shows persistence through symmetry-governed transformation. Primes show irreducibility under a declared operation. Infinite registries show that discovery and certification have stages.

Together these examples support one methodological proposal:

> Before accepting or rejecting a causal description, identify the frame that makes its entities and relations available. Then determine what that frame preserves, predicts, excludes, compresses, and leaves as residue when translated into another.

A Compact formal glossary

Term	Meaning in this paper
Process record	Declared carrier of states, samples, observations, events, or histories.
Frame	Domain, grain, selections, identity binder, event partition, ordering, causal relation, and observation map.
Identity binder	Rule grouping manifestations as one or more continuing entities.
Event partition	Rule deciding which changes count as bounded occurrences.
Causal graph	Directed event graph generated under a declared frame.
Bridge	Explicit translation between frames or causal graphs.
Invariant	Declared structure preserved under selected transformations.
Residue	Structure a translation, closure, compression, or comparison fails to preserve.
Registry time	Stage at which an object, witness, or explanation becomes available.
Frame collision	Undeclared mixing of commitments from different frames.

B Minimal rule-contract schema

A causal or relational rule should disclose:

- rule identity and branch provenance;
- participating roles;
- domain and frame;
- preconditions and trigger;
- transformation or response;
- propagation method and latency;
- conserved quantities or invariants;
- forbidden outcomes;
- uncertainty and tolerance;
- observation products;
- exceptions and priority rules.

A finite-propagation rule, for example, prohibits receiver response before channel arrival. A narrative salience rule may alter an agent's attention without acting as a physical force. Keeping these contracts typed allows both rules to coexist without ontological collapse.

C Formal claim register

Established mathematics used

- polygonal convergence to circle circumference;
- exponential limits and the differential equation $\dot{x} = rx$;
- Euler's formula and identity;

- Sophie Germain's factorization;
- Noether's variational theorem under stated hypotheses;
- prime irreducibility in $\mathbb{N}$;
- p-adic valuation and norm;
- finite signal propagation in the selected physical model;
- standard evolutionary population reasoning and the distinction between present function and historical cause.

Framework definitions introduced or consolidated

- descriptive-causal frame;
- frame-generated causal graph;
- identity binder;
- causal residue register;
- typed causal relation set;
- frame collision.

Modeling interpretations

- Eulerian closure as causal compression;
- Germain factorization as a prototype of relational decomposition and reattachment;
- prime artifacts as registry-specific visibility of base-independent irreducibility;
- registry stages as epistemic time in causal reconstruction;
- narrative teleology as coarse-grained compression of distributed process.

Open claims

- a general category of descriptive-causal frames;
- domain-independent quantitative residue metrics;
- obstruction theory for global causal reattachment;
- validated software demonstrations across physical, biological, and narrative domains.

D Source lineage within the EmergenceByDesign program

This paper consolidates and reorients material from prior laboratory records, especially:

- the corrected Lambda residue-accounting draft;
- the Registry Factorization Principle;

- the Primitive Semantic Registry Protocol;
- the Infinite Registry and Indexed-Residue Proof Engine;
- the Previs Studio causal-resolution notes;
- the Rope-and-Sand correction distinguishing primality from registry-visible prime artifacts.

The consolidation changes the hierarchy of those materials. Identity types do not form the paper's center. They serve as exemplars within the primary investigation of how descriptive frames construct causal subjects, event boundaries, temporal order, and permissible causal relations.

References

Darwin, Charles (1859). *On the Origin of Species*. John Murray.

Einstein, Albert (1905). "On the Electrodynamics of Moving Bodies". In: *Annalen der Physik* 17. English translations widely available, pp. 891–921.

Gould, Stephen Jay and Richard C. Lewontin (1979). "The Spandrels of San Marco and the Panglossian Paradigm: A Critique of the Adaptationist Programme". In: *Proceedings of the Royal Society of London. Series B* 205.1161, pp. 581–598.

Gouvêa, Fernando Q. (1997). *p-adic Numbers: An Introduction*. 2nd ed. Springer.

Kosmann-Schwarzbach, Yvette (2020). "The Noether Theorems in Context". In: *arXiv preprint arXiv:2004.09254*.

Mayr, Ernst (1961). "Cause and Effect in Biology". In: *Science* 134.3489, pp. 1501–1506.

National Institute of Standards and Technology (2026). *Digital Library of Mathematical Functions*. NIST online reference. Complex exponential and trigonometric functions; accessed August 2026.

Noether, Emmy (1971). "Invariant Variation Problems". In: *Transport Theory and Statistical Physics* 1.3. English translation by M. A. Tavel of Invariante Variationsprobleme (1918), pp. 186–207.

Pearl, Judea (2009). *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge University Press.

Semita, Lu (2026a). "Lambda Seems Legit! A Corrected Draft on the Lambda Principle as Residue Accounting Across Established Mathematical Obstruction Theorems". EmergenceByDesign laboratory draft.

— (2026b). "Previs Studio Concept: Causal-Resolution Layer Notes". EmergenceByDesign design record.

— (2026c). "Primitive Semantic Registry Protocol, Version 0.2". EmergenceByDesign protocol record.

— (2026d). "Registry Factorization Principle: From Closure to Transition". EmergenceByDesign working draft.

— (2026e). "The Infinite Registry and Indexed-Residue Proof Engine, Version 2". EmergenceByDesign laboratory notes record.